Week 4 Capstone
Phase 1 Close

Complete Architecture · Closing Open Items · Honest Scorecard · Phase 1 in One Page · The Resume Handoff

WEEK 4 · DAY 28	**Sunday, August 2 2026**	**Day 28 of 84 — PHASE 1 COMPLETE**

■■ **THE LAST SESSION BEFORE THE PAUSE.** Series resumes Monday 31 August 2026 at Day 29 / Week 5 (Agents & Tool Calling). Twenty-seven days, ~550 pages, one production system built layer by layer. Today is not a new topic — it is the session that determines whether any of it survives four weeks of not thinking about it.

What Today Is For

```
  1. ASSEMBLE — the whole architecture in one view, so you can see the
     SHAPE rather than 27 separate lessons

  2. CLOSE — the open items carried since Day 21. Two are one-line fixes
     that have been "next week" for eleven days.

  3. MEASURE — the honest final state, including what does NOT fit

  4. HAND OFF — an artifact written for someone who has forgotten
     everything, because on 31 August THAT PERSON IS YOU

THE PRINCIPLE FOR TODAY:
  A system you cannot resume is a system you have to REBUILD.
  The handoff is the DELIVERABLE, not a formality after the deliverable.
```

Sec	Topic	Coverage
Sec 1	Complete Architecture	Full request path, every day's contribution mapped
Sec 2	Closing Open Items	llm_query_role, golden-set gate, isolation tests
Sec 3	Honest Final State	Scorecard, accuracy budget, what I'd tell a reviewer
Sec 4	Phase 1 in One Page	All 27 days, the five generalizing ideas
Sec 5	The Resume Handoff	RESUME_BRIEF for 31 August, the handoff test
Sec 6	FAANG at the Pause	Google/Microsoft/Amazon/Meta/Netflix on handoffs

Sec 7	Architect's Lens	What transferred, what I'd do differently, honest state
Sec 8	Pause Checklist	Before closing, during the pause, on 31 August

Section 1 — The Complete Architecture

```
               +--------------------------------------+
               |         USER QUESTION                |
               +------------------+-------------------+
                                  |
   +------------------------------v-----------------------+
   |  GUARDRAIL: INPUT           D25                      |
   |  injection scan | PII tokenize | residency check     |
   +------------------------------+-----------------------+
                                  |
   +------------------------------v-----------------------+
   |  SEMANTIC CACHE             D24   ~18% hit -> return |
   |  threshold 0.97 | per-tenant namespace               |
   +------------------------------+-----------------------+
                                  | miss
   +------------------------------v-----------------------+
   |  ARTIFACT RESOLUTION        D23                      |
   |  prompt@ver | few-shot@ver | model@ver | retr-cfg@ver|
   +------------------------------+-----------------------+
                                  |
   +------------------------------v-----------------------+
   |  ORCHESTRATOR               D20    trace_id issued   |
   +------------------------------+-----------------------+
                                  |
+---------------------------------+--------------------------+
|  PHASE 1 - PARALLEL                                        |
|  +-----------------------+  +---------------------------+  |
|  | RETRIEVAL AGENT  8B   |  | LONG-TERM MEMORY    D19   |  |
|  | hybrid: dense+BM25    |  | past verified SQL as      |  |
|  | +RRF+MMR+rerank D17   |  | dynamic few-shot          |  |
|  | schema index     D16  |  +---------------------------+  |
|  | + document index D26  |                                 |
|  +-----------------------+      ~250 ms                    |
+---------------------------------+--------------------------+
                                  |
   +------------------------------v-----------------------+
   |  COMPLEXITY ROUTER          D24                      |
   |  tables | joins | hard markers | memory hit          |
   +-------+------------------------------+---------------+
           | simple                       | complex
+----------v----------+        +----------v----------+
| GENERATION  8B  D27 |        | GENERATION  70B     |
| fine-tuned candidate|        | CoT + few-shot  D18 |
+----------+----------+        +----------+----------+
           |  fails validation -> escalate |
           +--------------+---------------+
                          |
   +----------------------v-------------------------------+
   |  VALIDATION AGENT  8B       D20                      |
   |  EXPLAIN dry-run | faithfulness (sqlglot AST)   D18  |
   |  max 2 retries with corrected SQL                    |
   +----------------------+-------------------------------+
                          |
   +----------------------v-------------------------------+
   |  GUARDRAIL: OUTPUT      D22/D25    cheapest first    |
   |  read-only | PII cols | tenant scope | row cap |     |
   |  EXPLAIN plan cost                                   |
   +----------------------+-------------------------------+
                          |
   +----------------------v-------------------------------+
   |  EXECUTE   llm_query_role | RLS | statement_timeout  |
   +----------------------+-------------------------------+
                          |
   +----------------------v-------------------------------+
   |  RESPONSE  sql | rows | citations D18 |              |
   |            artifact_versions D23 | trace_id          |
   +----------------------+-------------------------------+
```

```
                          |
     +--------------------------v--------------------------------+
     |  ALWAYS-ON, OUT OF BAND                                    |
     |  audit log + hash chain D25 | cost attribution D24 |       |
     |  canary suite hourly D22 | drift detectors D22 |           |
     |  training data mining D27                                  |
     +------------------------------------------------------------+

FOUNDATIONS UNDERNEATH (Days 1-14 — not on the request path, load-bearing)
  D1-3   tokenization, embeddings, transformer   -> why context costs
  D4-5   training lifecycle, scaling laws, CoT   -> why CoT works
  D6     structured output                       -> why SQL parses
  D7     model selection                         -> why 70B/8B
  D8-10  quantization, LoRA, full fine-tuning    -> how D27 is possible
  D11-13 serving, evaluation, efficiency         -> why 900 q/hr
  D14    Text-to-SQL foundation + SPIDER         -> the benchmark
```

Section 2 — Closing the Open Items

Day 21's review named these. Day 22 repeated them. Day 25 ranked them TIER 1 — the controls that survive every application bug. They have been open for ELEVEN DAYS. Two are one-liners.

```
=== OPEN ITEM 1: DATABASE PRIVILEGE (Tier 1, ~20 minutes) =============
CREATE ROLE llm_query_role NOLOGIN;
GRANT USAGE ON SCHEMA analytics TO llm_query_role;
GRANT SELECT ON analytics.orders, analytics.customers,
                analytics.products, analytics.order_items
      TO llm_query_role;

-- The grant that matters is the one you DON'T write:
--   no INSERT/UPDATE/DELETE/TRUNCATE/DDL
--   no access to auth_tokens, user_credentials, audit_log

ALTER ROLE llm_query_role SET statement_timeout = '10s';
ALTER ROLE llm_query_role SET idle_in_transaction_session_timeout = '30s';

ALTER TABLE analytics.orders ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON analytics.orders
    FOR SELECT TO llm_query_role
    USING (tenant_id = current_setting('app.tenant_id')::text);
-- repeat per tenant-scoped table

VERIFY: connect as llm_query_role, run DELETE FROM analytics.orders LIMIT 1
        -> must fail with insufficient_privilege, NOT with a guardrail
        message. If a guardrail catches it first you tested the WRONG
        LAYER — disable the app guardrail and re-run to confirm the
        DATABASE refuses.

=== OPEN ITEM 2: GOLDEN-SET CI GATE (Day 23, ~1 day) =================
Wire ci/golden_gate.py as a REQUIRED status check on any PR touching:
    prompts/**  config/retrieval.yaml  config/models.yaml
    data/few_shot_bank.jsonl

The ENFORCEMENT is already built (Day 23): promote(stage="production")
raises PermissionError without a passing eval_run_id. What is missing is
only the CI wiring that PRODUCES that id.

VERIFY: open a PR removing the CoT instruction from the system prompt.
        -> the gate must BLOCK with an accuracy regression near -10pp.
        A GATE THAT HAS NEVER BLOCKED ANYTHING IS NOT KNOWN TO WORK.

=== OPEN ITEM 3: ISOLATION TESTS IN CI (Day 25, ~1 day) ==============
IsolationTestSuite -> required CI check. Five tests:
    retrieval | cache | memory | row-level enforcement | error messages

WHY THIS MATTERS MORE THAN IT LOOKS:
    Tenant isolation regressions are introduced by ORDINARY REFACTORS,
    not by attackers. Someone adds a convenience method that forgets a
    tenant_id parameter, tests pass, and a cross-tenant leak ships.
    This suite is the only thing between that refactor and a
    REPORTABLE BREACH.

VERIFY: temporarily remove the namespace from the semantic cache key.
        -> test_cache_isolation MUST fail. If it passes, the test is not
        exercising the path and is WORSE THAN NO TEST.
```

Status After Today — Honest Framing

Item	Status
llm_query_role + RLS	**SPECIFIED** — 20 min of SQL, no code change
golden-set CI gate	**SPECIFIED** — wiring only, logic already exists
IsolationTestSuite in CI	**SPECIFIED** — tests exist, need the required check
prompt caching deployment	STILL OPEN — largest remaining free win (~27%, zero accuracy cost)
PII log audit	STILL OPEN — one hour, will find real leaks (Day 25 Ex. 2)
get_sample_rows sanitizer	STILL OPEN — highest-risk tool still returns real rows

Today SPECIFIES the three Tier-1 items precisely enough to execute without re-deriving them. It does not EXECUTE them — that is a session with a live database, not a lesson. The specification is the deliverable that makes the execution a 20-minute task in September rather than a re-investigation.

Section 3 — The Honest Final State

```
CAPSTONE SCORECARD — Text-to-SQL System, Days 13-27

QUALITY
  Execution accuracy         82.8%    SLO > 82%          PASS (thin)
  Faithfulness               97.5%    SLO > 97%          PASS
  Syntax validity            98.5%    SLO > 98%          PASS
  Success rate               97.0%    SLO > 95%          PASS
    by difficulty: easy 95.9% | medium 88.5% | hard 73.2% | extra 57.1%

LATENCY
  p50                       2,180 ms
  p95                       3,740 ms  SLO < 4,000 ms     PASS
  p99                       5,920 ms  (mitigated, not blocking)

COST
  Per query              $0.002694   from $0.006300      -57.2%
  Throughput              1,240 q/hr from 900
  Monthly @ 500k queries      $1,347  from $3,150

ACCURACY BUDGET   <- THE NUMBER TO CARRY INTO SEPTEMBER
  Started                     2.5 pp   (84.5% vs 82% floor)
  Spent on cost optimization  1.7 pp   (Day 24)
  ----------------------------------
  REMAINING                   0.8 pp

  THIS IS THE BINDING CONSTRAINT ON THE ENTIRE SYSTEM.
  Day 27's generation fine-tune needed 1.6pp. IT DOES NOT FIT.
  Any future optimization must either fit in 0.8pp or first BUY headroom
  by improving accuracy somewhere.

SECURITY POSTURE
  Tier 1 (survives app bugs)          SPECIFIED, not deployed  <- the gap
  Tier 2 (survives model compromise)  DEPLOYED — 5 guardrails
  Tier 3 (reduces attack success)     DEPLOYED — fencing, detection
  Tier 4 (detection/forensics)        PARTIAL — audit designed,
                                      isolation tests not gating CI

OPERATIONAL MATURITY (Day 22 ladder)
  Level 0  deployed                   OK
  Level 1  full logging + trace_id    OK
  Level 2  canary suite + regression  OK  designed, hourly
  Level 3  guardrails + cost controls OK  app layer; DB layer specified
  Level 4  incidents -> eval cases    NOT DONE  <- the honest gap
```

The Three Things I Would Tell a Reviewer, Unprompted

```
1. THE ACCURACY BUDGET IS NEARLY SPENT AND THAT IS THE REAL RISK.
   82.8% against an 82% floor is 0.8pp of margin. One bad prompt change,
   one schema migration, one model version bump and the system is out of
   SLO. The correct next move is NOT another optimization — it is BUYING
   ACCURACY BACK. Day 21's failure triage identified two prompt fixes
   (set operations, HAVING placement) worth an estimated 3pp for two days
   of work. DO THAT BEFORE ANYTHING ELSE.

2. TIER 1 SECURITY IS SPECIFIED BUT NOT DEPLOYED.
   Every guardrail in this system is APPLICATION-LAYER. They are good
   guardrails. They are also ALL IN ONE PYTHON PROCESS, and a bug in that
   process removes ALL OF THEM SIMULTANEOUSLY. The database grant is 20
   minutes of work and it is the only control that survives that scenario.
```

```
   It has been open for eleven days. THAT is the finding.

3. WE HAVE NEVER RUN A ROLLBACK OR AN INCIDENT DRILL.
   Day 23 built rollback. Day 22 wrote the triage runbook. NEITHER HAS
   BEEN EXECUTED. Untested rollback is a HYPOTHESIS. The first real
   incident will be the test, and that is the worst possible time to
   discover the procedure has a gap.
```

Section 4 — Phase 1 in One Page

Day	Topic	The one thing that matters
1	Tokenization	Tokens are the billing unit and the context unit
2	Embeddings	Similarity is geometry; cosine over normalized vectors
3	Transformers	Attention is quadratic — why long context costs
4	Training lifecycle	Pre-train vs fine-tune vs prompt: different problems
5	Scaling laws + CoT	CoT is the cheapest accuracy in the whole curriculum
6	Structured output	Constrain the format, don't ask for it
7	Model selection	Weighted criteria, not "which is best"
8	Quantization	4x-8x memory reduction, minimal quality loss
9	LoRA / PEFT	Adapters make specialization economically viable
10	Full fine-tuning	Catastrophic forgetting is real and measurable
11	Serving	Continuous batching: 210 → 900 q/hr
12	Evaluation	Pick the metric that matches the failure mode
13	Efficiency	PagedAttention enables prefix sharing
14	Text-to-SQL foundation	Execution accuracy, not exact match
15	RAG foundations	Retrieval adds knowledge; nothing else does
16	Indexing pipelines	Fingerprints make re-indexing idempotent
17	Hybrid retrieval	Dense + BM25 + RRF + MMR + rerank
18	Generation	CoT + faithfulness + self-correction + citations
19	Agents & tools	Agent = loop; MAX_STEPS is a safety control
20	Multi-agent	Specialize by role → mix model sizes → cut cost
21	**Capstone**	Ablation justifies architecture: RAG = +53.5pp
22	LLMOps	Your failure mode is a 200 OK with a wrong answer
23	Versioning	A prompt is a model weight you can edit in a text editor
24	Cost	73% of input tokens are identical every request
25	Security	Assume injection succeeds; design for a compromised model
26	Multi-modal	An image is tokens; extract once, index forever
27	Fine-tuning	Doesn't add knowledge; buys cost and latency

```
THE FIVE IDEAS THAT GENERALIZE BEYOND THIS SYSTEM

  1. MEASURE BEFORE OPTIMIZING, ABLATE TO JUSTIFY.
     Day 21's ablation is the single most reusable artifact. It converts
```

```
   "we think retrieval helps" into "retrieval is worth 53.5pp and here
   is the number." Every architectural claim should have one.

2. CONTROLS BELONG OUTSIDE THE MODEL.
   Anything enforced by asking the model nicely is a HOPE. Anything
   enforced by a database grant, an AST check, or a routing decision is
   a CONTROL. This distinction survives every model generation.

3. THE FAILURE MODE IS SILENT SUCCESS.
   LLM systems return well-formed, confident, WRONG answers with no
   exception. Detection must be built DELIBERATELY — canary suites,
   drift detectors, abstention rates. Standard observability catches
   ONE failure class of FOUR.

4. BUDGETS ARE SHARED AND FINITE.
   Accuracy headroom is an ACCOUNT. Cost optimization draws on it.
   Fine-tuning draws on it. Track ONE balance or approve two changes
   that EACH fit and TOGETHER break the SLO.

5. VERSION EVERYTHING THAT AFFECTS OUTPUT.
   Prompts, few-shot banks, models, embedders, retrieval configs,
   guardrail thresholds, describers, datasets. Unversioned means
   unreproducible means undebuggable.
```

Section 5 — The Resume Handoff

> Written for someone who has forgotten everything. On 31 August, that person is YOU. This is the section to re-read first.

```
===================================================================
  RESUME BRIEF — read on Monday 31 August 2026 before Day 29
===================================================================

WHERE YOU STOPPED
  Day 28, Sunday 2 August 2026. Phase 1 complete: Weeks 1-4, Days 1-28.
  28 lesson PDFs in ~/Desktop/AI-ML/ and in the repo under week-NN/notes/.
  All Eagle Eye verified. Repo clean and pushed.

WHAT YOU BUILT
  A production Text-to-SQL system. Natural-language question in,
  validated SQL out, executed against a multi-tenant database, with
  citations. Built across Days 13-27 as ONE CONTINUOUS THREAD.

  Request path: guardrails -> semantic cache -> artifact resolution ->
  orchestrator -> parallel(retrieval, memory) -> complexity router ->
  generation(8B or 70B) -> validation -> output guardrails -> execute.

THE NUMBERS THAT DEFINE ITS STATE
  82.8% execution accuracy   (SLO floor 82% — 0.8pp of margin)
  p95 3,740 ms               (SLO 4,000 ms)
  $0.002694 per query        (down 57.2% from baseline)
  1,240 q/hr sustained

THE ONE CONSTRAINT TO REMEMBER
  ACCURACY BUDGET: 0.8pp REMAINING.
  Everything else follows from this. You cannot spend accuracy for cost
  or latency again until you buy some back.

FIRST THREE ACTIONS ON RESUME (in this order)
  1. BUY ACCURACY HEADROOM — Day 21's failure triage named two prompt
     fixes (nested set operations, HAVING vs WHERE placement) worth an
     estimated 3pp for ~2 days. This UNBLOCKS everything else.
  2. DEPLOY TIER 1 SECURITY — llm_query_role + RLS. 20 minutes of SQL.
     Specified in full in Day 28 Section 2.
  3. WIRE THE TWO CI GATES — golden set (Day 23) and IsolationTestSuite
     (Day 25). The logic EXISTS; only the CI wiring is missing.

WHAT IS SPECIFIED BUT NOT DEPLOYED
  - llm_query_role + row-level security        (Day 25/28)
  - golden-set CI gate wiring                  (Day 23)
  - IsolationTestSuite as required check       (Day 25)
  - prompt caching in production               (Day 24 — ~27%, FREE)
  - get_sample_rows sanitizer                  (Day 25/26)
  - PII audit across production logs           (Day 25)

WHAT WEEK 5 COVERS
  Agents & Tool Calling — Days 29-35, starting Monday 31 August 2026.
  You already built an agent on Day 19 (ReAct loop, tools, memory) and a
  multi-agent system on Day 20. Week 5 goes DEEPER: tool-calling
  patterns, agent frameworks, and the Tool-Calling Agent portfolio
  project. RE-READ DAYS 19-20 BEFORE DAY 29 — that is the foundation
  Week 5 builds on.
```

```
WHERE EVERYTHING LIVES
  Lessons:  ~/Desktop/AI-ML/WeekN_DayNN_*.pdf
  Repo:     git@github.com:irudi2021/ai-engineering-journey.git
  Schedule: SCHEDULE.md in the repo root — authoritative, locked
  Protocol: 4 steps per day — teach, Eagle Eye, PDF, push.
            FAANG rule: Meta/Apple/Amazon/Netflix/Google/Microsoft only.
===================================================================

THE HANDOFF TEST
  A handoff is adequate if a competent engineer who has NEVER SEEN the
  system can, FROM THIS DOCUMENT ALONE:
    1. State what the system does and its measured performance
    2. Name the binding constraint and why it binds
    3. List the next three actions in priority order
    4. Find every artifact without asking anyone

  If any of those requires a CONVERSATION, the handoff has FAILED —
  and in four weeks THERE IS NOBODY TO HAVE THAT CONVERSATION WITH.
```

Section 6 — FAANG: What Happens at the Pause Point

Google — Design Docs as the Durable Artifact

Google's engineering culture is well known for the design document preceding significant work, and the property that makes the practice durable is that the document OUTLIVES THE CONTEXT in which it was written. A design doc records not only the decision but the ALTERNATIVES CONSIDERED and the reasons they were rejected — exactly the information that evaporates from memory first and is most expensive to reconstruct. Day 21's ablation table and Day 28's scorecard serve the same function here: months from now, "why is retrieval in this architecture" is answerable with a NUMBER rather than a recollection.

Microsoft — Runbooks Written for the Person Who Wasn't There

Microsoft's operational guidance for Azure services consistently emphasizes runbooks that assume NO PRIOR CONTEXT — the on-call engineer at 3am may never have seen the service before. The discipline transfers directly to a project pause: the resume brief in Section 5 is a runbook whose reader happens to be your FUTURE SELF, and the same rule applies — anything requiring tribal knowledge to act on is NOT WRITTEN DOWN YET. The test is not whether the author can follow it, but whether someone WITHOUT the author's memory can.

Amazon — The Written Narrative Over the Presentation

Amazon's practice of replacing slide decks with written six-page narratives rests on the argument that PROSE FORCES THE AUTHOR to make a complete, connected case in a way that bullet points allow you to avoid. A pause forces the same discipline for the same reason: the summary tables in this lesson are NAVIGATIONAL, but Section 3's "three things I would tell a reviewer" is where the actual state of the system is stated honestly, IN SENTENCES, including what does not work. Bullets let you list features; PROSE MAKES YOU ADMIT the accuracy budget is nearly spent.

Meta — Bounded Milestones Over Continuous Grind

Meta's engineering cadence organizes work into defined milestones with explicit review points rather than open-ended continuous effort, on the reasoning that a CHECKPOINT is where scope gets re-examined against what was actually LEARNED. A 28-day pause is an unusually clean version of that checkpoint. The system as it stands has a known constraint that only became visible once Day 24's cost work and Day 27's fine-tune were measured against THE SAME BUDGET — a discovery that a continuous grind would likely have deferred until something broke in production.

Netflix — Context Over Control

Netflix's engineering culture emphasizes giving people the CONTEXT to make good decisions rather than the PROCESS to follow. For a paused project, context is the ENTIRE DELIVERABLE: the resume brief exists precisely so that the decision "what do I do first on 31 August" can be made correctly by someone with NO MEMORY of the reasoning that produced the current state. Section 5's ordering — buy accuracy headroom BEFORE deploying anything else — is a decision that only makes sense with the budget context attached, which is why the constraint is stated BEFORE the action list rather than after it.

Section 7 — Architect's Lens

```
WHAT 28 DAYS OF BUILDING ONE SYSTEM ACTUALLY TAUGHT

  The lessons that transferred were NEVER THE FRAMEWORKS.
  vLLM, Qdrant, sqlglot, LoRA — all replaceable, all WILL be replaced.
  What transferred:

    - Measure before optimizing; ablate to justify
    - Put controls OUTSIDE the model
    - Assume the SILENT failure, not the loud one
    - Track budgets as SHARED accounts
    - Version anything that affects output
    - Prefer the REVERSIBLE change (prompt) over the durable one (weights)

  A curriculum organized around ONE CONTINUOUS SYSTEM produces this.
  Twenty-eight disconnected tutorials would not have — because the
  accuracy budget only became visible when Day 24's cost work and Day
  27's fine-tune were measured against THE SAME NUMBER.

THE MOST VALUABLE HABIT FROM PHASE 1
  EAGLE EYE. Not the script — the DISCIPLINE.
  "Never declare a gap without verification; never declare coverage
  without confirmation." That habit applied to a curriculum is a checking
  routine. Applied to a PRODUCTION SYSTEM it is the difference between an
  engineer who says "it works" and one who says "it works, HERE IS THE
  MEASUREMENT, and here is WHAT I COULD NOT VERIFY."

WHAT I WOULD DO DIFFERENTLY ACROSS DAYS 1-28
  1. Deploy Tier 1 security on Day 21, when it was FIRST IDENTIFIED.
     Eleven days open on a 20-minute task is the clearest PROCESS FAILURE
     in Phase 1, and it happened because it was never the most
     INTERESTING thing available to work on.
  2. Track the accuracy budget from Day 21 rather than DISCOVERING it on
     Day 24. The constraint existed the whole time; only the ACCOUNTING
     was late.
  3. Run one rollback drill. IT IS STILL NOT DONE.

THE HONEST STATE OF THE SYSTEM
  PRODUCTION-CAPABLE, NOT PRODUCTION-DEPLOYED.
  Architecture: sound and measured.
  Security:     correct design, Tier 1 not yet enforced at the database.
  Operations:   instrumented, drills never run.
  Economics:    understood, with 0.8pp of margin.

  That is a GOOD PLACE TO PAUSE. It is NOT a good place to ship.
  Knowing the difference is what Phase 1 was for.

FOUR WEEKS OFF IS NOT A COST
  The pause lands exactly on the Week 4 boundary — 28 days, no split
  week. Everything is measured, committed, and written down for a reader
  with no memory. RESUMING IS READING ONE PAGE, not reconstructing four
  weeks of reasoning.
```

Section 8 — The Pause Checklist

Before You Close This Session

```
# 1. Confirm the repo is clean and pushed
#      git status && git log --oneline -1
# 2. Confirm all 28 PDFs are present in ~/Desktop/AI-ML/
# 3. Read SCHEDULE.md once — confirm 31 August, Week 5, Day 29
# 4. Read the RESUME BRIEF (Section 5) once, SLOWLY.
#      If any line requires context you have not written down, ADD IT NOW.
```

Optional During the Pause — Highest Value First

```
# 1. Deploy llm_query_role + RLS    20 min · closes the Tier 1 gap
# 2. Run the PII log audit          1 hour · Day 25 Exercise 2
# 3. Deploy prompt caching          1 day  · ~27%, ZERO accuracy cost
# 4. The two prompt fixes           2 days · buys back ~3pp of budget
#
# NONE of this is required. ALL of it makes 31 August easier.
# If you do EXACTLY ONE THING, do #1 — it is twenty minutes and it is the
# only control that survives an application bug.
```

On 31 August, Before Day 29

```
# 1. Read RESUME BRIEF (Section 5)
# 2. Re-read Day 19 (Agents and Tool Use) and Day 20 (Multi-Agent) —
#    Week 5 builds DIRECTLY on both
# 3. Run one canary evaluation to confirm nothing drifted while idle
# 4. Say "Day 29 Week 5 Start"
```

Phase 1 Complete

Days	28 of 84 (33%)
Weeks	4 of 12
PDFs	28, all Eagle Eye verified
Pages	~550
System	Production Text-to-SQL — 82.8% accuracy, $0.002694/query
Repo	Clean, pushed, SCHEDULE.md locked
Resumes	**Monday 31 August 2026 · Day 29 · Week 5 · Agents & Tool Calling**

■■ PHASE 1 CLOSED — SEE YOU ON 31 AUGUST 2026

Twenty-eight days. One system. Measured, documented, and written down for a reader with no memory. Week 5 resumes with Agents & Tool Calling, building on Days 19–20. Read the resume brief first.